

ETH: METHOD III DATABASE

Jüttner, D., Labarthe, S., Prastalo, N.

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CODEBOOK

Identification

- `match_id` - Unique numeric identifier for each match
- `tournament_id` - Unique identifier for the World Cup tournament
- `year` - Year of the World Cup tournament
- `group_name` - Name of the qualification group (e.g., “Group A”)
- `group_stage` - Group stage identifier
- `stage_name` - Stage of the competition (“group stage”)
- `match_date` - Date of the match (YYYY-MM-DD format)
- `country_name` - Name of the focal country/team
- `country_code` - ISO 3-letter country code (e.g., “FRA”, “GER”, “BRA”)
- `team_id` - Unique numeric identifier for the focal team
- `opponent_name` - Name of the opposing country/team
- `opponent_id` - Unique numeric identifier for the opponent
- `opponent_code` - ISO 3-letter code of the opponent
- `venue` - Match venue from focal team’s perspective (“home” or “away”)

Football variables

- `goals_for` - Number of goals scored by focal team
- `goals_against` - Number of goals conceded by focal team
- `goal_difference` - Goals for minus goals against ($\text{goal_difference} = \text{goals_for} - \text{goals_against}$)
- `win` - Binary indicator for win (1 = win, 0 = otherwise)
- `draw` - Binary indicator for draw (1 = draw, 0 = otherwise)
- `loss` - Binary indicator for loss (1 = loss, 0 = otherwise)
- `points` - Points earned in match (3 for win, 1 for draw, 0 for loss)
- `matches_played` - Total number of matches played by team in group stage
- `wins` - Total number of wins in group stage
- `draws` - Total number of draws in group stage
- `losses` - Total number of losses in group stage
- `rank` - Team’s rank within the qualification group (1 = highest)
- `teams_qualifying` - Number of teams that qualify from each group (1 for year ≤ 1970 , 2 otherwise)
- `qualified` - Binary indicator for World Cup qualification (1 = qualified, 0 = not qualified)
- `min_qualifier_points` - Minimum points among teams that qualified in the group
- `max_non_qualifier_points` - Maximum points among teams that did not qualify in the group
- `points_gap` - Difference between `min_qualifier_points` and `max_non_qualifier_points`
- `tie_at_cutoff` - Binary indicator for tie at qualification cutoff (1 = tie exists, 0 = otherwise)
- `treatment` - Binary treatment variable (1 = qualified, 0 = not qualified)
- `points_diff` - Running variable for RDD: `points_gap` if qualified, `-points_gap` if not qualified

Military variables

- `bicc_milper` - Military Personnel Index (BICC, 1990-2023)
- `bicc_milexp` - Military Expenditure Index (BICC, 1990-2023)
- `bicc_gmi` - Global Militarization Index (BICC, 1990-2023)
- `wdi_expmil` - Military expenditure as percentage of GDP (World Bank, 1960-2024)

Demographic variables

- `wdi_pop` - Total population (World Bank, 1960-2024)
- `wdi_pop_log` - Natural logarithm of total population
- `wdi_popurb` - Urban population (World Bank, 1960-2024)
- `wdi_birth` - Birth rate per 1,000 people (World Bank, 1960-2024)
- `wdi_death` - Death rate per 1,000 people (World Bank, 1960-2024)
- `wdi_mortinf` - Infant mortality per 1,000 live births (World Bank, 1960-2024)
- `wdi_lifexp` - Life expectancy at birth in years (World Bank, 1960-2023)
- `wpp_medianage` - Median age as of July 1 (UN World Population Prospects, 1950-2024)
- `wpp_sexratio` - Sex ratio (males per 100 females) (UN World Population Prospects, 1950-2024)

Economic variables

- `wdi_export` - Exports of goods and services as percentage of GDP (World Bank, 1960-2024)
- `wdi_import` - Imports of goods and services as percentage of GDP (World Bank, 1960-2024)
- `wdi_gdpcapcon2015` - GDP per capita in constant 2015 US dollars (World Bank, 1960-2024)
- `wdi_gdpgr` - GDP growth rate (annual percentage) (World Bank, 1961-2024)
- `wdi_area` - Land area in square kilometers (World Bank, 1961-2023)

Political variables

- `vdem_polyarchy` - Electoral democracy index (VDEM, 1946-2024)
- `ht_colonial` - Colonial origin indicator (HT, 1946-2025)
- `atop_number` - Number of alliances (ATOP, 1946-2025)
- `v2paseatshare` - Seat share of nationalist parties in parliament (VDEM)

Notes

- NA - Missing values are present for some variables due to data availability limitations
- Time coverage varies by variable as indicated in parentheses above

DATABASE REPLICATION

Nationalism (share of seat in parliament)

```
#install.packages("remotes")
remotes::install_github("vdeminstitute/vdemdata")

library(vdemdata)
library(dplyr)

vp <- vparty %>%
  rename(country_code = country_text_id) %>%
  select(country_code, year, v2paseatshare) %>%
  group_by(country_code, year) %>%
  summarise(
```

```

v2paseatshare = mean(v2paseatshare, na.rm = TRUE),
.groups = "drop"
)

```

QOG

```

library(readr)

url_data <- "https://raw.githubusercontent.com/SimonLabarthe/Method-3/main/qog.rds"
qog <- read_rds(url_data) %>%
  rename(country_name = cname,
          country_code = ccodealp) %>%
  select(
    country_name,
    country_code,
    year,

    bicc_milper,      # Military Personnel Index (i = 162, t=1990-2023)
    bicc_milexp,      # Military Expenditure Index (i = 162, t=1990-2023)
    bicc_gmi,         # Global Militarization Index (i = 162, t=1990-2023)
    wdi_expmil,       # Military expenditure % of GDP (i = 163, t=1960-2024))

    wdi_pop,          # Total Population (i = 192, t = 1960-2024)
    wdi_popurb,       # Urban Population (i = 192, t = 1960-2024)

    wdi_birth,        # Birth Rate (per 1,000) (i = 192, t = 1960-2024)
    wdi_death,        # Death Rate (per 1,000) (i = 192, t = 1960-2024)
    wdi_mortinf,      # Infant Mortality (per 1,000 live births) (i = 192, t = 1960-2024)
    wdi_lifexp,       # Life Expectancy at birth (per 1,000 live births) (i = 192, t = 1960-2023)

    wpp_medianage,    # Median Age as of 1 July (WPP) (i = 192, t = 1950-2024)
    wpp_sexratio,     # Sex Ratio: males per 100 females (WPP) (i = 192, t = 1950-2024)

    wdi_export,       # Exports of goods & services (% of GDP) (i = 173, t = 1960-2024)
    wdi_import,       # Imports of goods & services (% of GDP) (i = 173, t = 1960-2024)
    wdi_gdpcapcon2015, # GDP per capita (constant 2015 US dollar) (i = 191, t = 1960-2024)
    wdi_gdpgr,        # GDP growth (annual %) (i = 191, t = 1961-2024)

    wdi_area,         # Land area in sq. km (i = 191, t = 1961-2023)

    vdem_polyarchy,   # Democracy level (i = 175, t = 1946-2024)
    ht_colonial,       # Colonial Origin (i = 198, t = 1946-2025)

    atop_number,      # Number of Alliances (i = 198, t = 1946-2025)
  ) %>%
  mutate(
    wdi_pop_log = log(wdi_pop)
  )

summary(qog)

```

FIFA World Cup

```
remotes::install_github("jfelstul/worldcup")

library(tidyverse)
library(worldcup)
library(lubridate)

data("matches")
data("tournaments")
data("teams")
data("qualified_teams")

qualification_matches <- matches %>%
  filter(stage_name == "group stage") %>%
  mutate(
    year = as.numeric(str_extract(tournament_name, "\\d{4}"))
  ) %>%
  filter(year %% 2 == 0) # Only males

home_team_matches <- qualification_matches %>%
  select(
    tournament_id,
    year,
    match_id,
    match_date,
    group_name,
    group_stage,
    stage_name,
    team_id = home_team_id,
    country_name = home_team_name,
    team_code = home_team_code,
    opponent_id = away_team_id,
    opponent_name = away_team_name,
    opponent_code = away_team_code,
    goals_for = home_team_score,
    goals_against = away_team_score,
    win = home_team_win,
    draw,
    loss = away_team_win
  ) %>%
  mutate(
    points = case_when(
      win == 1 ~ 3,
      draw == 1 ~ 1,
      TRUE ~ 0
    ),
    goal_difference = goals_for - goals_against,
    venue = "home"
  )

away_team_matches <- qualification_matches %>%
  select(
    tournament_id,
```

```

year,
match_id,
match_date,
group_name,
group_stage,
stage_name,
team_id = away_team_id,
country_name = away_team_name,
team_code = away_team_code,
opponent_id = home_team_id,
opponent_name = home_team_name,
opponent_code = home_team_code,
goals_for = away_team_score,
goals_against = home_team_score,
win = away_team_win,
draw,
loss = home_team_win
) %>%
mutate(
  points = case_when(
    win == 1 ~ 3,
    draw == 1 ~ 1,
    TRUE ~ 0
  ),
  goal_difference = goals_for - goals_against,
  venue = "away"
)

monadic_matches <- bind_rows(home_team_matches,
                             away_team_matches) %>%

  arrange(year,
           group_name,
           country_name,
           match_date)

qualification_standings <- monadic_matches %>%
  group_by(year, group_name, team_id, country_name, team_code) %>%
  summarise(
    matches_played = n(),
    wins = sum(win, na.rm = TRUE),
    draws = sum(draw, na.rm = TRUE),
    losses = sum(loss, na.rm = TRUE),
    goals_for = sum(goals_for, na.rm = TRUE),
    goals_against = sum(goals_against, na.rm = TRUE),
    goal_difference = sum(goal_difference, na.rm = TRUE),
    points = sum(points, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(year, group_name, desc(points), desc(goal_difference), desc(goals_for))

qualification_standings <- qualification_standings %>%
  group_by(year, group_name) %>%
  mutate(
    rank = row_number(),

```

```

teams_qualifying = case_when(
  year <= 1970 ~ 1, # Earlier tournaments had 1 qualifier per group
  year <= 2300 ~ 2, # Later tournaments had 2
  TRUE ~ 2
),
qualified = ifelse(rank <= teams_qualifying, 1, 0)
) %>%
ungroup()

tight_races <- qualification_standings %>%
  rename(country_code = team_code) %>%
  group_by(year, group_name) %>%
  mutate(
    min_qualifier_points = min(points[qualified == 1], na.rm = TRUE),
    max_non_qualifier_points = max(points[qualified == 0], na.rm = TRUE),
    points_gap = min_qualifier_points - max_non_qualifier_points,
    tie_at_cutoff = any(qualified == 1 & points == max_non_qualifier_points) |
      any(qualified == 0 & points == min_qualifier_points)
  ) %>%
  ungroup() %>%
  filter(
    (qualified == 1 & points == min_qualifier_points) |
    (qualified == 0 & points == max_non_qualifier_points)
  ) %>%
  mutate(
    treatment = case_when(
      qualified == 1 ~ 1,
      qualified == 0 ~ 0
    ),
    points_diff = ifelse(
      qualified == 1, points_gap, -points_gap
    )
  )
)

```

Merge

```

library(countries)

data <- tight_races %>%
  left_join(qog, by = c("country_code", "year")) %>%
  left_join(vp, by = c("country_code", "year")) %>%
  filter(!is.na(treatment),
    points_diff != "Inf")

```